

# 1.INTRODUCTION

The Periyar University NSS Portal is a web-based application designed to digitize and simplify the management of National Service Scheme (NSS) activities across different institutional levels. In the existing system, processes such as volunteer enrollment, event management, and report submission are carried out manually, leading to inefficiency, errors, and poor data management. This project introduces a centralized platform that connects administrators, colleges, and NSS units, ensuring better coordination and transparency.

The system is structured into multiple functional modules including Authentication & Secure Access, Administrator Command, College Administration, Unit Operations & Volunteer Management, Student Enrolment & Demographic, Automated Document & Reporting, and Event Management & Field Activity modules. Each module is designed to handle specific operations, ensuring a well-organized and scalable system architecture.

The system follows a role-based access structure, where Admin, College, and Unit users have specific permissions and responsibilities. It is developed using modern technologies to ensure performance and scalability. The frontend is built using React with Vite for a fast and responsive interface, while the backend uses Node.js and Express to handle server-side operations. MongoDB is used for efficient data storage, and communication between components is handled through RESTful APIs. Secure authentication is implemented to protect user data, along with additional features like email notifications.

The application provides functionalities such as digital volunteer enrollment, event creation and tracking, collaborative participation, and report generation. Overall, the system improves efficiency, reduces manual work, and offers a reliable solution for managing NSS activities in a structured and effective manner.

## **1.2.SYSTEM SPECIFICATION :**

### **1.2.1.HARDWARE CONFIGURATION:**

- **Processor** : Dual-Core Processor or higher (e.g., Intel Core i3 or equivalent) with a clock speed of 2.0 GHz or higher.
- **RAM** : Minimum 4 GB of RAM.
- **Storage** : Minimum 512 MB of free disk space.
- **Network** : Stable internet connection for cloud database connection(MongoDB atlas)

The selected hardware configuration provides sufficient performance to run development tools like Node.js, React, and database services smoothly. A minimum of 4 GB RAM and a dual-core processor ensures efficient multitasking during coding and testing. Stable internet connectivity is essential for accessing cloud services like MongoDB Atlas and Cloudinary.

### **1.1.2.SOFTWARE SPECIFICATION:**

- **Operating System** : Windows 10 or 11, Linux (Ubuntu 16.04 or later).
- **Framework** : Node.js with Express.js for backend.
- **Libraries** : React.js for frontend.
- **Package Manager** : Node Package Manager ( npm ) for managing dependencies.
- **Database** : MongoDB ( Atlas for Cloud Database ).
- **API Testing** : Postman.
- **Cloud storage** : Cloudinary ( For storing event brochures )
- **Version Control** : Git and GitHub.
- **Code Editor** : Visual Studio Code (or) CodeSandbox( for online coding ).
- **Browser** : Latest versions of Chrome, Firefox, Microsoft Edge or Brave.

The chosen MERN stack enables efficient full-stack development using a single language (JavaScript), improving development speed and consistency. Node.js and Express.js support scalable backend development, while React.js provides a dynamic user interface. Additionally, MERN applications can be easily deployed on cloud platforms, making the system flexible and accessible. Tools like MongoDB Atlas, GitHub, and Postman ensure reliable database management, version control, and API testing.

## **2.SYSTEM STUDY**

### **2.1.EXISTING SYSTEM**

#### **2.1.1. DESCRIPTION**

The existing system for managing NSS activities is mostly manual and paper-based. Processes like volunteer enrollment, event management, and report submission are handled using physical forms such as Annexure-A. Data is stored in files or registers, making it difficult to organize and retrieve information.

In addition, the Existing NSS webpage <https://www.periyaruniversity.ac.in/nss.php> mainly provides downloadable links for forms and displays a few past event images, but lacks interactive features or a centralized digital system for managing activities. Communication between administrators, colleges, and units is not centralized, leading to delays and confusion.

#### **2.1.2. DRAWBACKS**

- **Manual Process:** Most NSS activities such as enrollment, attendance, and report preparation are handled using paper-based methods, increasing manual effort.
- **Time Consumption:** The overall process is slow, as tasks like data entry, verification, and report generation require significant time.
- **Human Errors:** Manual handling of data often leads to mistakes in recording, duplication, and loss of important information.
- **Poor Data Management:** Maintaining records in physical files or basic formats makes it difficult to store, organize, and retrieve data efficiently.
- **Lack of Centralization:** There is no single unified platform to manage all activities, leading to scattered and uncoordinated operations.
- **Low Transparency:** Administrators and higher authorities have limited visibility into the activities of colleges and units.
- **Tracking Issues:** Monitoring volunteer participation, event progress, and outcomes becomes challenging without a proper system.

## **2.2.PROPOSED SYSTEM**

### **2.2.1. DESCRIPTION**

The proposed system is a web-based **Periyar University NSS Portal** designed to automate and simplify NSS activities through a centralized platform. It replaces manual processes like volunteer enrollment, event management, and reporting with a digital solution. The system uses role-based access control for Admin, College, and Unit users to ensure secure and organized operations. It allows easy storage, retrieval, and management of data in real time. The platform also supports collaborative events and improves communication between different levels. Automated notifications and structured report generation enhance efficiency. Overall, the system reduces manual work, minimizes errors, and improves transparency.

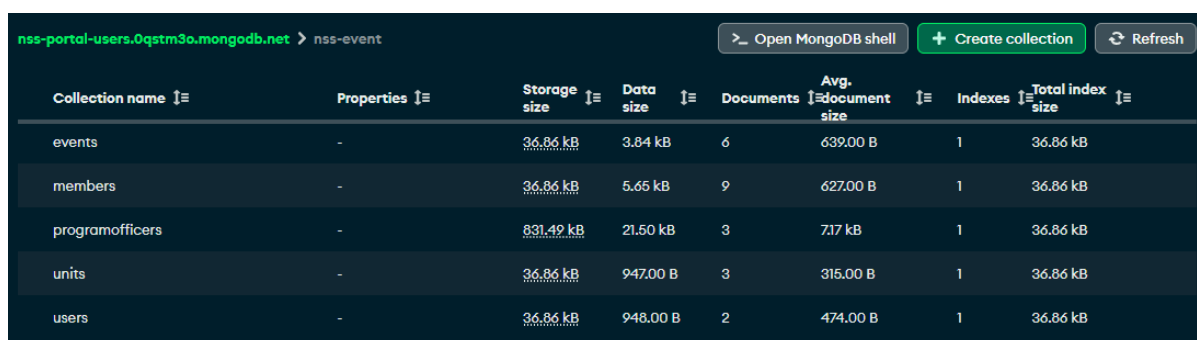
### **2.2.2. FEATURES**

- **Digital Enrollment:** Provides an online system for registering NSS volunteers with detailed personal, and activity-related information, replacing paper-based forms.
- **Role-Based Access Control:** Ensures secure and structured access for different users such as Admin, College, and Unit, with specific permissions for each role.
- **Event Management:** Enables users to create, update, and manage NSS events, including scheduling, descriptions, and tracking progress.
- **Collaborative Events:** Allows multiple NSS units to participate and coordinate in a single event, improving teamwork and resource sharing.
- **Centralized Data Storage:** Maintains all data in a unified database, making it easy to access, update, and manage information efficiently.
- **Automated Notifications:** Sends real-time alerts and updates to users regarding new events, changes, and report submissions.
- **Report Generation:** Automatically generates structured reports for events and volunteer participation, reducing manual effort.
- **Data Export Options:** Supports exporting data in formats such as PDF, Excel, and CSV for administrative and official purposes.

## 3. SYSTEM DESIGN AND DEVELOPMENT

### 3.1.DATABASE DESIGN

The database design of the **Periyar University NSS Portal** is structured to efficiently manage data related to users, units, volunteers, events, and program officers. It follows a flexible schema approach to handle different types of data while maintaining proper relationships between entities. Each component in the system is interconnected, ensuring smooth data flow and consistency. The design supports scalability, allowing the system to handle increasing data over time. It also ensures efficient storage, retrieval, and updating of records.



The screenshot shows the MongoDB Atlas interface for the 'nss-portal-users.0qstm3o.mongodb.net' database. The 'nss-event' database is selected. The interface includes buttons for 'Open MongoDB shell', 'Create collection', and 'Refresh'. A table lists the collections in the database:

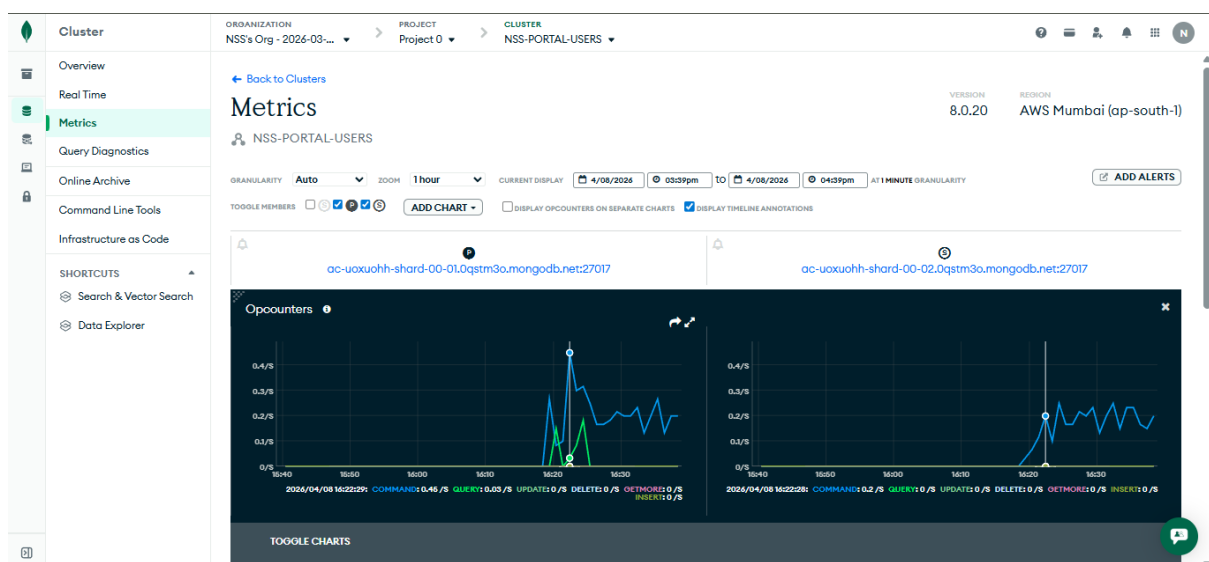
| Collection name | Properties | Storage size | Data size | Documents | Avg. document size | Indexes | Total index size |
|-----------------|------------|--------------|-----------|-----------|--------------------|---------|------------------|
| events          | -          | 36.86 kB     | 3.84 kB   | 6         | 639.00 B           | 1       | 36.86 kB         |
| members         | -          | 36.86 kB     | 5.65 kB   | 9         | 627.00 B           | 1       | 36.86 kB         |
| programofficers | -          | 831.49 kB    | 21.50 kB  | 3         | 717 kB             | 1       | 36.86 kB         |
| units           | -          | 36.86 kB     | 947.00 B  | 3         | 315.00 B           | 1       | 36.86 kB         |
| users           | -          | 36.86 kB     | 948.00 B  | 2         | 474.00 B           | 1       | 36.86 kB         |

The **Member**, **Unit**, and **User** models form the core structure of the system. The Member model stores detailed volunteer information, including personal, academic, and enrollment details, and is linked to specific units and colleges. The Unit model manages NSS units, their members, and associated activities within a college. The User model represents institutional-level data, including login credentials and organizational details, and maintains relationships with multiple units.

The **Event** and **Program Officer** models handle operational and administrative functionalities. The Event model stores complete event details such as schedule, venue, participants, collaborators, and final reports. It also supports tracking attendance and post-event outcomes. The Program Officer model maintains details of NSS coordinators, including their personal and professional information. Together, these models ensure efficient data management, reduce redundancy, and support the overall functioning of the system.

| Table(Collection) Name       | Purpose                                                                        |
|------------------------------|--------------------------------------------------------------------------------|
| User (College)               | Manages institution login, overall units, events, and adopted villages         |
| AdoptingVillages (Sub-table) | Stores details of villages adopted by the college for NSS activities           |
| Unit                         | Represents an NSS unit under a college, manages members and events             |
| Member                       | Stores student details including personal info and NSS enrolment data          |
| Event                        | Manages event details like schedule, venue, participants, and unit association |
| ProgramOfficer               | Stores details of NSS officers who manage and guide units                      |
| Collaborators (Sub-table)    | Handles collaboration between different units for an event                     |
| Report (Sub-table)           | Stores post-event report details like participants count, outcome, and files   |

MongoDB is used as the database management system to store and manage all application data efficiently. It provides a flexible, document-based structure that allows easy handling of complex and dynamic data. As we use the MongoDB Atlas (Cloud) Its scalability and high performance support the smooth functioning of the Periyar University NSS Portal .



## 3.2.INPUT DESIGN

Basic Information

EVENT NAME \*

Enter event name

EVENT DESCRIPTION \*

EVENT CATEGORY \*

Select Category

☒

THIS IS A SINGLE-DAY EVENT

DATE \*

mm/dd/yyyy

TIME FROM

--:-- --

TIME TO

--:-- --

VENUE \*

Enter event venue

Media & Documents

EVENT IMAGES

Choose Files

No file chosen

Upload Selected Images

No images selected or uploaded yet.

EVENT BROCHURE (PDF)

Choose File

No file chosen

Upload Brochure

Collaboration

COLLABORATING UNITS (OPTIONAL)

Select a Unit to Invite

Add

Selected units will receive an invitation to collaborate on this event.

Cancel

Create Event

## Register Program Officer Pro-form-a

**Main Details**

NAME OF PROGRAM OFFICER
OFFICER-4

DESIGNATION
Professor

DEPARTMENT
Computer Science

UNIT (OPTIONAL)
Unassigned

**Personal Details**

PROFILE IMAGE
Choose File sample-profile.png

CURRENT PHOTO

Download Form PDF
Close
Register Officer

The input design of the **Periyar University NSS Portal** focuses on providing a user-friendly and consistent interface for data entry. All input fields such as text boxes, dropdowns, date pickers, and file uploads are designed with a clean and modern style using rounded corners, proper spacing, and readable fonts. Interactive states like hover and focus are included to guide users during data entry. The system also applies input validation to ensure accuracy and reduce errors. This improves overall usability and ensures reliable data collection.

The application uses structured forms for different modules to simplify complex inputs. The **Event Creation Form** allows users to enter details like event name, category, date, venue, and collaborators in an organized manner. The **Volunteer Enrollment Form (Annexure-A)** is divided into sections such as personal, academic, physical, and skills to reduce user complexity. Advanced UI elements like icons, placeholders, and grouped inputs further enhance clarity. Overall, the input design ensures efficient, accurate, and smooth user interaction.



3.3.OUTPUT DESIGN



Registered Colleges2 total

Click a row to enter the college dashboard

| CODE | INSTITUTION NAME | UNITS | EVENTS | ACTIONS |
|------|------------------|-------|--------|---------|
| 888  | Sample Institute | 2     | 5      | Remove  |
| 333  | Sample2          | 1     | 1      | Remove  |

Admin DashboardSYSTEM

Periyar University NSS Portal

+ Register CollegeSign Out

Events CalendarView All --

April 2026

| Sun | Mon | Tue | Wed | Thu | Fri | Sat |
|-----|-----|-----|-----|-----|-----|-----|
|     |     |     | 1   | 2   | 3   | 4   |
| 5   | 6   | 7   | 8   | 9   | 10  | 11  |
| 12  | 13  | 14  | 15  | 16  | 17  | 18  |
| 19  | 20  | 21  | 22  | 23  | 24  | 25  |
| 26  | 27  | 28  | 29  | 30  |     |     |

Click on dates with colored dots to view events

Quick Actions

Manage All Events

Program Officers

Student Records

Total Events6

Upcoming Events3

Address: 9/197 Udhayam Nagar, Kamalapuram, Omalur Salem

ACADEMIC & ETI

Educational Qualification: MCA

ETI Training Completed: Yes

Seminars / Workshops / Courses:

• NIL

|    |                                   |                        |
|----|-----------------------------------|------------------------|
| 11 | MOBILE NUMBER NO.                 | 9988 / 10000           |
| 12 | E-mail ID                         | dinesh.m2003@gmail.com |
| 13 | Aadhaar Card No.                  | 456789123456           |
| 14 | Date/Year of Enrolment            | 2026-03-27             |
| 15 | Cultural Talents (Please specify) | Folk Dance             |
| 16 | Hobbies                           | Football, Photography  |

Declaration

I DineshKumarM hereby agree to obey all the rules and regulations of National Service Scheme and work imbibing the spirit of the Scheme.

The output design of the **Periyar University NSS Portal** focuses on presenting data in a clear, structured, and visually appealing manner. The system provides dashboards with graphical representations such as charts and key metrics to help administrators easily analyze NSS activities. Visual elements like charts and calendars are used to display event distribution, institutional performance, and daily schedules. These outputs improve decision-making by providing real-time insights and better understanding of data.

The system also supports professional reporting and data export functionalities. It generates detailed PDF reports for completed events, including participation details, outcomes, and supporting images. Additionally, data can be exported in formats like Excel for offline use and official submissions. Automated email notifications are used to communicate updates and alerts to users. The system also produces standardized documents such as the Annexure-A enrollment form. Overall, the output design ensures effective data presentation, reporting, and communication.

### **3.4.CODE DESIGN**

The code design of the **Periyar University NSS Portal** follows a well-structured and modular architecture based on modern software engineering principles. The application is developed using the MERN stack, ensuring scalability, flexibility, and efficient handling of dynamic data. The frontend is built using a component-based approach, where reusable components such as forms, modals, dashboards, and input elements improve code reusability and maintainability. State management is handled using Context API, enabling smooth data flow across components, while routing is implemented with proper protection mechanisms to restrict unauthorized access.

The backend is designed following RESTful API architecture. Each API endpoint is structured with proper request handling and standardized responses, along with effective error handling using try-catch mechanisms. Security is a key aspect of the design, implemented through JWT-based authentication and role-based access control to manage Admin, College, and Unit privileges. Sensitive data such as passwords are securely handled using encryption techniques. The system also integrates third-party services for functionalities like email notifications, media storage, and data visualization. Overall, the code design ensures a secure, scalable, and maintainable system that supports efficient development and future enhancements.

frontend/

```
|— src/
|   |— components/    # All UI modules & Dashboards
|   |— context/       # State management (User/Theme)
|   |— assets/        # Global images & icons
|   └— App.jsx        # Main Routing & Entry point
```

backend/

```
|— server.js          # Core server logic & API Routes
|— middleware/        # Auth & Security checks
└— .env               # Secure configuration
```

### **3.5.SYSTEM DEVELOPMENT**

## **Requirement Analysis**

This phase involves understanding the needs and objectives of the NSS Portal. The system requirements are gathered by identifying the functionalities such as member management, unit handling, and activity tracking. Both functional and non-functional requirements are clearly defined to ensure the system meets user expectations.

## **System Design**

In this phase, the overall architecture of the system is planned. The application is designed using the MERN stack, ensuring a clear separation between frontend and backend. Database schema, user interface layout, and system workflows are structured to provide efficiency, scalability, and ease of use.

## **Implementation (Development)**

During implementation, the actual coding of the system is carried out. The frontend is developed using React.js to create an interactive user interface, while the backend is built using Node.js and Express.js to handle server-side operations. MongoDB is used to store and manage data efficiently.

## **Testing**

The system is thoroughly tested to ensure all functionalities work correctly. Different types of testing such as unit testing, integration testing, and user testing are performed. Errors and bugs are identified and fixed to improve system reliability and performance.

## **Deployment**

After successful testing, the system is deployed to a production environment (netlify for frontend and render for backend). This makes the NSS Portal accessible to users. Proper configuration and hosting are ensured so that the application runs smoothly in real-time usage.

## **Maintenance**

Maintenance is an ongoing process where the system is monitored and updated regularly. Any issues faced by users are resolved, and new features or improvements are added to enhance performance and usability over time.

### **3.5.1. MODULE DESCRIPTION**

#### **1. Authentication & Secure Access Module**

- **Role-Based Access Control (RBAC):** Implements specialized login gateways for three distinct user levels: System Administrator, College Coordinator, and NSS Unit Head.
- **JWT Security Integration:** Ensures data integrity using JSON Web Tokens (JWT) for all API requests, providing end-to-end encryption for session management.
- **Aesthetic UI Persistence:** Includes a specialized Theme Toggle system that dynamically switches between Light and Night modes with state persistence.

The image displays three distinct login interfaces side-by-side, each with a light blue header bar and a white main content area. The first interface on the left is for the 'SYSTEM ADMINISTRATOR', featuring a blue icon and the title 'Admin Sign In' with the subtitle 'Periyar University NSS Portal'. It includes fields for 'USERNAME' (with placeholder 'Enter admin username') and 'PASSWORD' (with placeholder 'Enter password' and a toggle icon), a blue 'Sign In' button, and a 'Back to Home' link. The middle interface is for the 'NSS UNIT', featuring a green icon and the title 'Unit Sign In' with the subtitle 'Access your NSS unit dashboard'. It includes fields for 'COLLEGE CODE' (placeholder 'e.g. 507'), 'UNIT CODE' (placeholder 'e.g. PUNSS...'), and 'PASSWORD' (placeholder 'Enter unit password' and a toggle icon), a blue 'Sign In' button, and a 'Back to Home' link. The third interface on the right is for the 'ORGANIZATION', featuring a blue icon and the title 'Organization Sign In' with the subtitle 'Access your college NSS dashboard'. It includes fields for 'USERNAME' (placeholder 'Enter your username') and 'PASSWORD' (placeholder 'Enter your password' and a toggle icon), a blue 'Sign In' button, and a 'Back to Home' link.

## 2. Administrator Command Module (Centralized Oversight)

- **Institutional Monitoring:** Provides a bird's-eye view of all registered colleges, their organizational structures and the events.
- **Master Student Registry:** A searchable database of all students across the region, featuring multi-layered filtering by community, college, and department.
- **Officer Management:** Centralized tracking and management of Program Officers' registration and performance metrics.

System Administrator

## All Events Management

[Back to Dashboard](#)

SEARCH EVENTS

Search by name, code, or college...

FILTER BY CATEGORY

All Categories

Total Events

6

Filtered Results

6

Categories

6

## Events List

| EVENT CODE            | EVENT NAME     | CATEGORY          | COLLEGE              | UNIT                | DATE(S)    | VENUE                                                      | ACTIONS                      |
|-----------------------|----------------|-------------------|----------------------|---------------------|------------|------------------------------------------------------------|------------------------------|
| NSSEVTPUNSSAM88801001 | Sample Event   | Tree Plantation   | Sample Institute 888 | unit1 PUNSSAM88801  | 2026-04-08 | Periyar university Campus                                  | <a href="#">View Details</a> |
| NSSEVTPUNSSAM33301001 | Sample Event-2 | Awareness Program | Sample2 333          | unit2 PUNSSAM33301  | 2026-04-07 | Seminar Hall, Dept of Computer Science, Periyar University | <a href="#">View Details</a> |
| NSSEVTPUNSSAM88801002 | Sample Event-3 | Cleanliness Drive | Sample Institute 888 | unit1 PUNSSAM88801  | 2026-04-09 | University Campus                                          | <a href="#">View Details</a> |
| NSSEVTPUNSSAM88802001 | Sample Event   | Seminar           | Sample Institute 888 | Unit 2 PUNSSAM88802 | 2026-03-25 | Seminar Hall, Dept of Computer Science, Periyar University | <a href="#">View Details</a> |
| NSSEVTPUNSSAM88801003 | event-test     | Sports            | Sample Institute 888 | unit1 PUNSSAM88801  | 2026-04-10 | Ground                                                     | <a href="#">View Details</a> |
| NSSEVTPUNSSAM88801004 | mail test      | Workshop          | Sample Institute 888 | unit1 PUNSSAM88801  | 2026-04-12 | seminar hall                                               | <a href="#">View Details</a> |

## All Program Officers

Admin Access

[Dashboard](#)

SEARCH BY NAME

Enter officer name...

FILTER BY COLLEGE

College name or code...

| ID         | OFFICER NAME | COLLEGE (CODE)       | UNIT         | DEPARTMENT | CONTACT    | ACTION                                    |
|------------|--------------|----------------------|--------------|------------|------------|-------------------------------------------|
| NSSPO88803 | Officer 3    | Sample Institute 888 | Unassigned   | CS         | 9876543210 | <a href="#">View</a> <a href="#">Edit</a> |
| N/A        | officer 3    | Sample Institute 888 | PUNSSAM88801 | Chemistry  | 9876543210 | <a href="#">View</a> <a href="#">Edit</a> |
| N/A        | Officer 2    | Sample2 333          | PUNSSAM33301 | CS         | 9876543210 | <a href="#">View</a> <a href="#">Edit</a> |

## All Students

Admin Access

[Dashboard](#)

SEARCH (NAME OR REG NO)

Enter student name or reg no...

FILTER BY COLLEGE

College name or code...

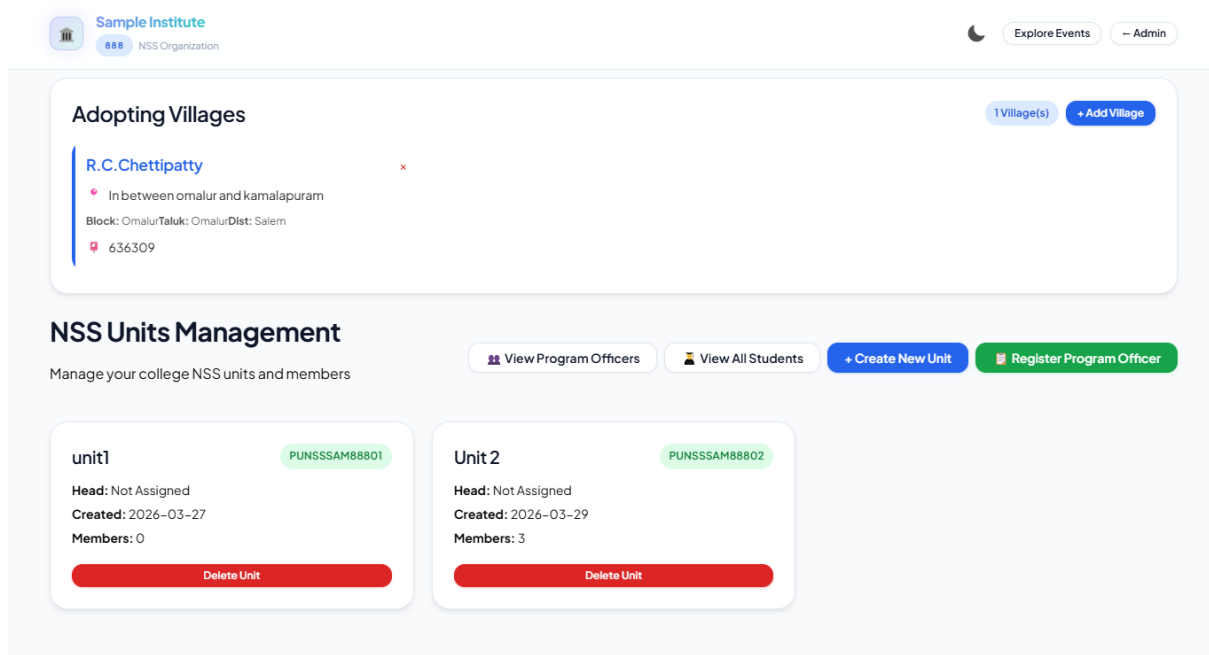
FILTER BY COMMUNITY

All Communities

| REG NO  | NAME         | COLLEGE (CODE)       | UNIT         | DEPT | COMMUNITY | BATCH       | ACTION                           |
|---------|--------------|----------------------|--------------|------|-----------|-------------|----------------------------------|
| 22EC045 | PriyaS       | Sample Institute 888 | PUNSSAM88802 | MCA  | SC        | 2022 - 2026 | <a href="#">✓ View Enrolment</a> |
| 22ME012 | DineshKumarM | Sample Institute 888 | PUNSSAM88802 | MBA  | MBC       | 2022 - 2026 | <a href="#">✓ View Enrolment</a> |
| 22CS001 | KarthikeyanR | Sample Institute 888 | PUNSSAM88802 | CS   | OBC       | 2022 - 2026 | <a href="#">✓ View Enrolment</a> |

### 3. College Administration Module (Institutional Dashboard)

- **NSS Unit Orchestration:** Facilitates the creation and management of up to six NSS units per institution.
- **Program Officer Recruitment:** Manages the appointment of officers, ensuring unique leadership per unit through intelligent synchronization logic.
- **Village Adoption System:** A dedicated interface for managing rural outreach data, capturing location-specific details like Block, Taluk, and District for adopted villages.



### 4. Unit Operations & Volunteer Management Module

- **Local Unit Control:** Enables Unit Heads to manage their specific group of assigned volunteers independently.
- **Volunteer Lifecycle Tracking:** Oversees everything from the initial enrolment of students to their active participation in social services.
- **Unit-specific Analytics:** Generates participation summaries and activity logs for internal unit evaluation.

Unit 2  
PUNSSSAM88802  
Sample Institute

College Dashboard

### Unit Details

UNIT CODE  
**PUNSSSAM88802**

UNIT NAME  
**Unit 2**

UNIT HEAD  
**Officer 3**

CONTACT  
**9876543210**

EMAIL  
**bsam53888@gmail.com**

### Quick Actions

Create Events

Manage events, volunteers and more from here.

Explore Events

Explore events created by other units.

### Unit Members

Manage student volunteers

Excel Export
UPLOAD VIA EXCEL ⓘ
+ Add Member

| <input type="checkbox"/> | S.NO | NAME         | REG NO  | DEPARTMENT | BATCH       | CONTACT    | ACTION                                             |
|--------------------------|------|--------------|---------|------------|-------------|------------|----------------------------------------------------|
| <input type="checkbox"/> | 1    | KarthikeyanR | 22CS001 | CS         | 2022 - 2026 | 9876543210 | <div>✓ Edit</div> <div>Delete</div> <div>PDF</div> |
| <input type="checkbox"/> | 2    | PriyaS       | 22EC045 | MCA        | 2022 - 2026 | 9123456789 | <div>✓ Edit</div> <div>Delete</div> <div>PDF</div> |
| <input type="checkbox"/> | 3    | DineshKumarM | 22ME012 | MBA        | 2022 - 2026 | 9988776655 | <div>✓ Edit</div> <div>Delete</div> <div>PDF</div> |

## 5. Student Enrolment & Demographic Module

- **Digital Enrolment Workflow:** A comprehensive digital form system that captures student bios, academic data, and contact information.
- **Demographic Analysis:** Features the specialized "Community Filtering" system to analyze volunteer diversity based on caste and community categories (OC, BC, MBC, SC, ST).
- **Record Sorting:** Implements advanced sorting algorithms to organize student records by institution and unit numbers.

### Add New NSS Volunteer

#### Basic Details

|                                                               |                                                                       |
|---------------------------------------------------------------|-----------------------------------------------------------------------|
| NAME *<br><input type="text" value="Full Name"/>              | REGISTER NUMBER *<br><input type="text" value="Registration Number"/> |
| DEPARTMENT *<br><input type="text" value="e.g. CSE"/>         | COURSE *<br><input type="text" value="e.g. B.E. / B.Tech"/>           |
| BATCH FROM (YEAR) *<br><input type="text" value="e.g. 2022"/> | BATCH TO (YEAR) *<br><input type="text" value="e.g. 2026"/>           |
| COMMUNITY *<br><input type="text" value="General"/>           | BLOOD GROUP *<br><input type="text" value="Select Blood Group"/>      |



|                 |                        |
|-----------------|------------------------|
| DATE OF BIRTH * | MOBILE PHONE NO. *     |
| mm/dd/yyyy      | 10-digit mobile number |

---

**Enrolment Details**

|                             |                              |
|-----------------------------|------------------------------|
| SEX *                       | FATHER/GUARDIAN'S NAME *     |
| Male                        | Enter father/guardian's name |
| FATHER/GUARDIAN'S CONTACT * | HEIGHT (CM) *                |
| 10-digit mobile number      | e.g. 165                     |
| WEIGHT (KG) *               | AADHAAR NUMBER *             |
| e.g. 60                     | 12-digit Aadhaar number      |
| EMAIL ID *                  | DATE OF ENROLMENT            |
|                             |                              |

|                                |                      |
|--------------------------------|----------------------|
| ENTER EMAIL ADDRESS            | DATE OF ENROLMENT    |
| Enter email address            | 04/08/2026           |
| CULTURAL TALENTS               | HOBBIES              |
| e.g. Singing, Dancing          | e.g. Reading, Sports |
| UNIVERSITY NAME                |                      |
| Enter full residential address |                      |

[Download Enrolment PDF](#)
[Cancel](#)
[Add & Enrol](#)

## 6. Event Management & Field Activity Module

- **Outreach Documentation:** Allows coordinators to schedule events, camps, and awareness programs.
- **Multimedia Reporting:** Supports high-resolution image uploads for event galleries and detailed activity report submission for validation.

**Explore Events**  
Discover what's happening in your unit and college

[Back to Dashboard](#)

**My Unit Events** [+ Add Event](#)

**Seminar** NSSEVTPUNSSAM88802001 2026-03-25

**Sample Event**  
sample  
Time: 10:00 - 17:00  
Venue: Seminar Hall, Dept of Computer Science, Periyar University

[Delete](#)  
[Edit Report](#)

**College Events**

**Tree Plantation** NSSEVTPUNSSAM88801001 2026-04-08

**Sample Event**  
sample event for testing event module  
Time: 10:00 - 17:00  
Venue: Periyar university Campus

**Cleanliness Drive** NSSEVTPUNSSAM88801002 2026-04-09

**Sample Event-3**  
Sample Event  
Time: 10:00 - 17:00  
Venue: University Campus

**Sports** NSSEVTPUNSSAM88801003 2026-04-10

**event-test**  
test  
Time: 10:00 - 17:00  
Venue: Ground

**Workshop** NSSEVTPUNSSAM88801004 2026-04-12

**mail test**  
hello  
Time: 10:00 - 17:00  
Venue: seminar hall

**Other Events**

**Awareness Program** NSSEVTPUNSSAM33301001 2026-04-07

**Sample Event-2**  
Sample testing event  
Time: 10:00 - 15:00  
Venue: Seminar Hall, Dept of Computer Science, Periyar University

## 7. Automated Document & Reporting Module

- **Dynamic PDF Engine:** Uses JS-driven templates to instantly generate official Government-standard Pro-forma and Enrolment forms.
- **Bulk Data Export:** Features a one-click "Export to Excel" functionality for all administrative lists, facilitating easy offline record-keeping and audit trails.
- **Bulk Data Import:** Provides a one-click "Import from Excel" feature that allows administrators to upload pre-formatted Excel sheets using a given template, enabling the addition of multiple student records at once quickly and efficiently.

### NOMINATION OF NEW PROGRAMME OFFICER TO LOOK AFTER THE NSS UNIT IN THE COLLEGE SAMPLE INSTITUTE

Name: Officer 3

Designation: Professor

Department: CS

Unit Assigned:



#### PERSONAL DETAILS

Date of Birth: 2004-10-19

Community: OBC

Email: bsam53888@gmail.com

Mobile: 9876543210

Block: Kamalapuram

Taluk: Omalur

District: Salem

Pincode: 636309

Date of Appointment: 2020-07-19

Teaching Experience: 5

Address: 9/197 Udhayam Nagar, Kamalapuram, Omalur Salem

#### ACADEMIC & ETI

Educational Qualification: MCA

ETI Training Completed: Yes

Seminars / Workshops / Courses:

• NIL

#### GENERAL

NSS Experience:

• NIL

Special Talents / Skills:

• Dancer

### ANNEXURE - A NSS VOLUNTEER ENROLMENT (AT UNIT LEVEL)

Name of the College: Sample Institute

Address: .....

Name of the University: Anna University

Unit No: PUNSSSAM88801

Affix passport size photo

|    |                                   |                                   |
|----|-----------------------------------|-----------------------------------|
| 1  | Name of the Volunteer             | DineshKumarM                      |
| 2  | Father/Guardian's Name            | Marimuthu D                       |
| 3  | Father/Guardian's Phone No.       | 9977665544                        |
| 4  | Sex                               | Male [✓] Female [ ]               |
| 5  | Date of birth                     | 2003-11-08                        |
| 6  | Year & Branch in which studying   | B.E. - MBA                        |
| 7  | Community                         | General [ ] SC [ ] ST [ ] OBC [ ] |
| 8  | Residential Address               | 78, Gandhi Road, Salem - 636001   |
| 9  | Blood Group                       | A+                                |
| 10 | Height in 168 (cm)                | Weight 70 kg                      |
| 11 | Mobile Phone No.                  | 9988776655                        |
| 12 | E-mail ID                         | dinesh.m2003@gmail.com            |
| 13 | Aadhaar Card No.                  | 456789123456                      |
| 14 | Date/Year of Enrolment            | 2026-03-27                        |
| 15 | Cultural Talents (Please specify) | Folk Dance                        |
| 16 | Hobbies                           | Foothall, Photography             |

#### Declaration

I **DineshKumarM** hereby agree to obey all the rules and regulations of National Service Scheme and work imbuing the spirit of the Scheme.

Date: 2026-03-27

Place:

Signature of the applicant

### Unit Members

Manage student volunteers

Search members...

Excel Export

UPLOAD VIA EXCEL

+ Add Member

| <input type="checkbox"/> | S.NO | NAME         | REG NO  | DEPARTMENT | BATCH       | CONTACT    | ACTION                                                            |
|--------------------------|------|--------------|---------|------------|-------------|------------|-------------------------------------------------------------------|
| <input type="checkbox"/> | 1    | KarthikeyanR | 22CS001 | CS         | 2022 - 2026 | 9876543210 | <a href="#">✓ Edit</a> <a href="#">Delete</a> <a href="#">PDF</a> |
| <input type="checkbox"/> | 2    | PriyaS       | 22EC045 | MCA        | 2022 - 2026 | 9123456789 | <a href="#">✓ Edit</a> <a href="#">Delete</a> <a href="#">PDF</a> |
| <input type="checkbox"/> | 3    | DineshKumarM | 22ME012 | MBA        | 2022 - 2026 | 9988776655 | <a href="#">✓ Edit</a> <a href="#">Delete</a> <a href="#">PDF</a> |

## 4. TESTING AND IMPLEMENTATION

Testing is an important phase in the development of the **Periyar University NSS Portal** to ensure that the system works correctly and efficiently. It is carried out to identify and fix errors before deployment. Different types of testing are performed to validate system functionality, performance, and security. The testing process ensures that all modules operate as expected under various conditions. Overall, it helps in delivering a reliable and user-friendly application.

### **Functional Testing**

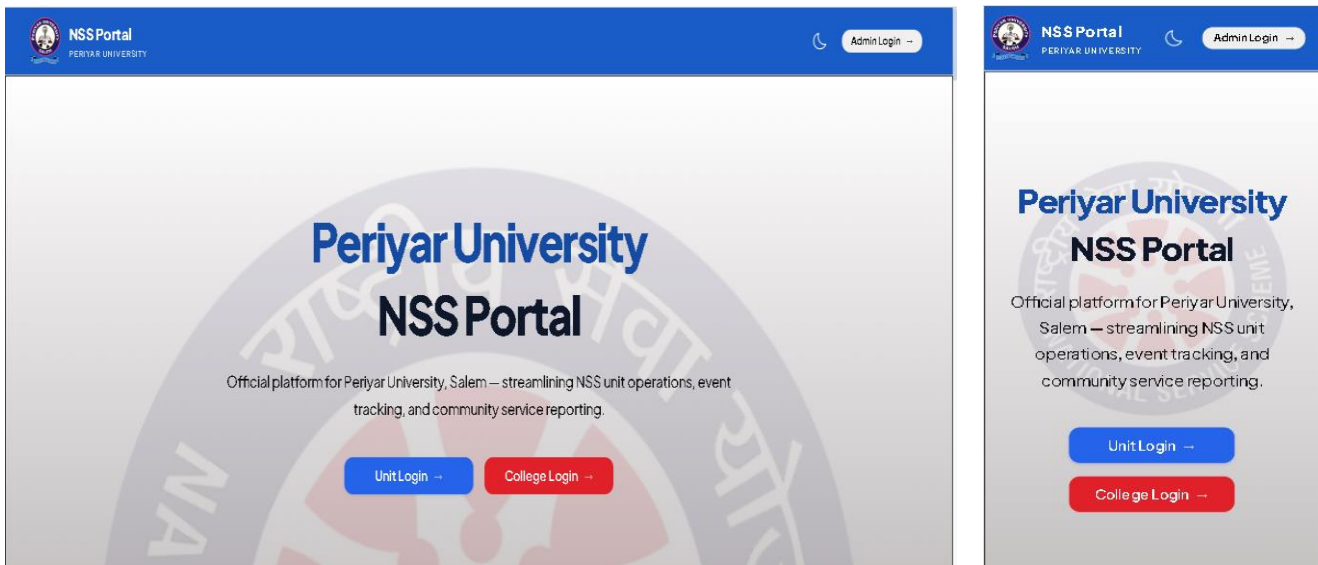
Functional testing is performed to verify that all core features of the system work as expected. The user registration process is tested to ensure unique credentials and secure password handling. Unit creation and management functionalities are validated based on defined constraints. The complete event lifecycle, including creation, updates, participation, and report submission, is thoroughly tested. This ensures that each module performs its intended function correctly.

### **Integration Testing**

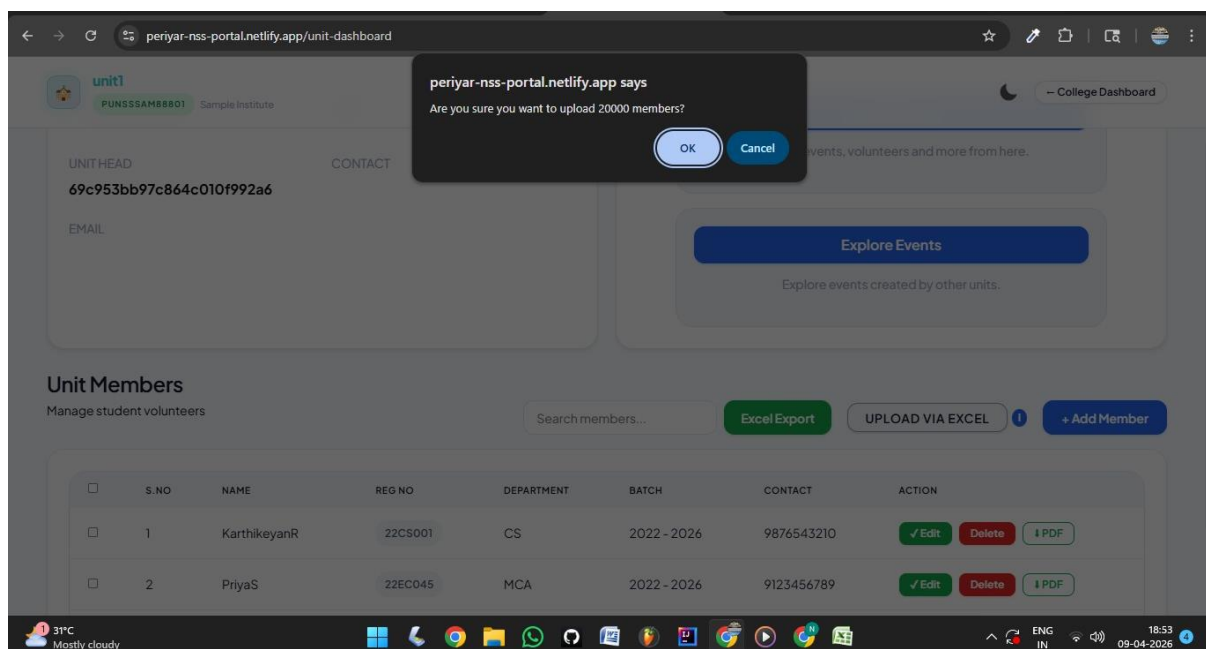
Integration testing ensures smooth interaction between different modules and external services. The communication between frontend and backend APIs is tested for accurate data transfer. Email notification services are verified to ensure alerts are sent correctly to users. Data export features such as Excel and PDF generation are tested to confirm proper formatting and data integrity. This testing guarantees that all integrated components work together seamlessly.

### **UI/UX Testing**

UI/UX testing is conducted to ensure the system is user-friendly and visually consistent. The application is tested across different devices such as desktops, mobiles, and tablets to verify responsiveness. User interface elements like forms, buttons, and dashboards are checked for proper alignment and usability. Theme support, including light and dark modes, is also tested for readability and consistency. This improves overall user experience and accessibility.



Performance testing is carried out to evaluate the system’s responsiveness and stability under different conditions. The application is tested with multiple users and large datasets to ensure smooth operation. Loading time, data processing speed, and system responsiveness are monitored. Optimizations are applied to maintain fast performance even with increased usage. This ensures reliability in real-world scenarios.



The screenshot shows a web application interface for the Periyar University NSS Portal. At the top, there is a navigation bar with the university logo, the name 'unit1', and a 'College Dashboard' link. Below this is a table listing student records. Each row contains a checkbox, a student ID (e.g., 20006), a name (e.g., DineshKumarM), a course code (e.g., 22ME012), a course name (e.g., MBA), a duration (2022 - 2026), a contact number (e.g., 9988776655), and three action buttons: 'Edit', 'Delete', and 'PDF'. The table is paginated, showing records 20001 to 20013 of 20013 students. The current page is 401, with previous and next buttons and a range of page numbers (397, 398, 399, 400, 401).

| Student ID | Name         | Course Code | Course Name | Duration    | Contact Number | Actions               |
|------------|--------------|-------------|-------------|-------------|----------------|-----------------------|
| 20006      | DineshKumarM | 22ME012     | MBA         | 2022 - 2026 | 9988776655     | [Edit] [Delete] [PDF] |
| 20007      | Sarath       | 22CS001     | CS          | 2022 - 2026 | 9876543210     | [Edit] [Delete] [PDF] |
| 20008      | Sam          | 22EC045     | MCA         | 2022 - 2026 | 9123456789     | [Edit] [Delete] [PDF] |
| 20009      | Deepak       | 22ME012     | MBA         | 2022 - 2026 | 9988776655     | [Edit] [Delete] [PDF] |
| 20010      | Gowtham      | 22CS008     | CS          | 2022 - 2026 | 9876543210     | [Edit] [Delete] [PDF] |
| 20011      | Mohammed     | 22EC049     | MCA         | 2022 - 2026 | 9123456789     | [Edit] [Delete] [PDF] |
| 20012      | Vinoth       | 22ME030     | MBA         | 2022 - 2026 | 9988776655     | [Edit] [Delete] [PDF] |
| 20013      | Vignesh      | 22ME048     | MBA         | 2022 - 2026 | 9988776655     | [Edit] [Delete] [PDF] |

The system was evaluated by importing a large dataset of around 20,000 student records at once. Although the data was successfully uploaded, a performance issue occurred while displaying the student information due to the large volume of data.

To resolve this issue, pagination was implemented, which limits the number of records displayed per page. This significantly improved performance and ensured smooth data retrieval and user experience. The solution effectively handled large-scale data without affecting system efficiency.

## **Security Testing**

Security testing is implemented to protect user data and system integrity. Authentication mechanisms are tested to ensure only authorized users can access the system. Token-based security is verified for protected routes and sensitive operations. Data encryption techniques are checked to safeguard user credentials. This ensures the system remains secure against unauthorized access and threats.

## **IMPLEMENTATION AND DEPLOYMENT**

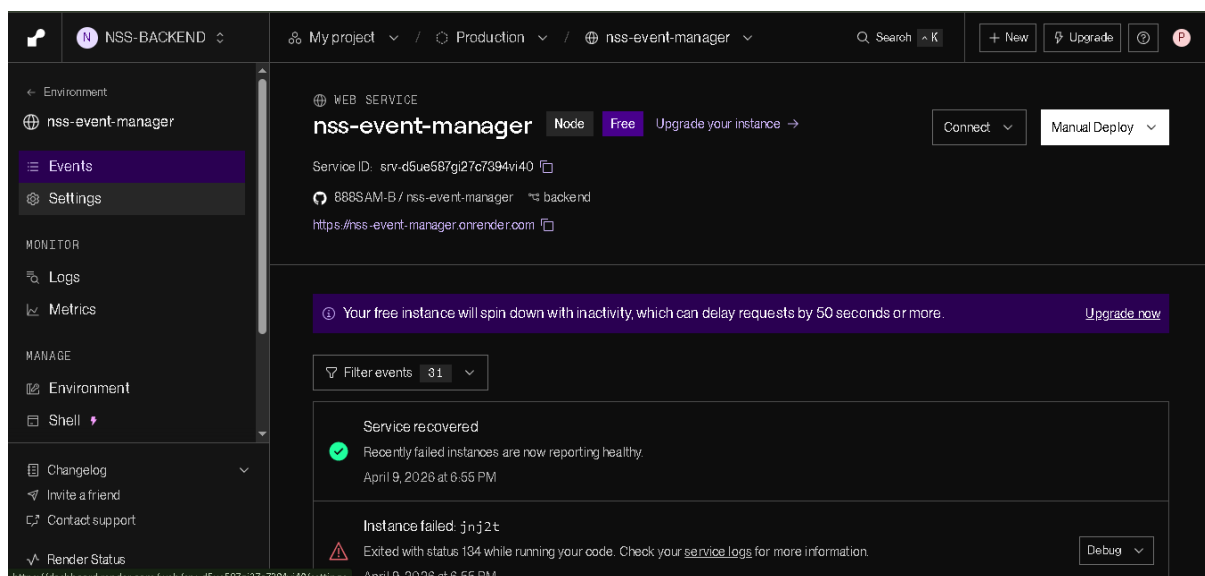
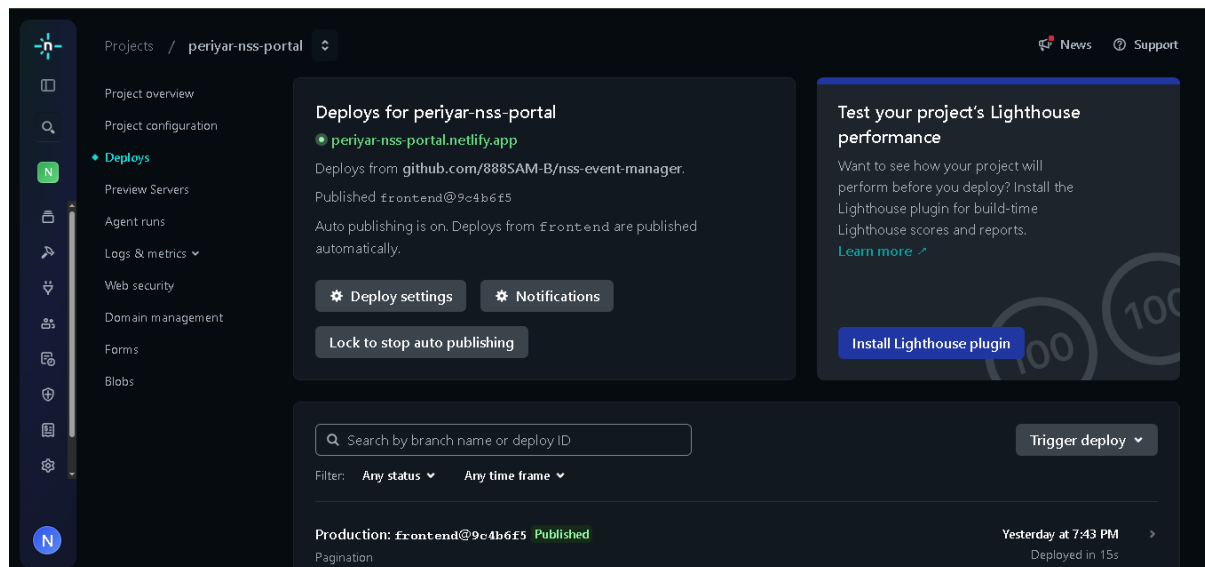
### **IMPLEMENTATION PHASE**

The implementation phase involves the development and integration of all modules of the **Periyar University NSS Portal**. During this phase, the system is built using a structured approach where frontend, backend, and database components are developed and connected. All

features such as user authentication, volunteer management, event handling, and reporting are implemented based on the system design. Proper coding standards and validation techniques are followed to ensure accuracy and efficiency. Continuous testing is also carried out during this phase to identify and fix errors.

## DEPLOYMENT PHASE

The Periyar University NSS Portal was deployed using modern cloud platforms to ensure accessibility and reliability. The frontend was hosted on Netlify, while the backend services were deployed using Render for efficient server-side operations. The database was managed using MongoDB Atlas, ensuring secure and scalable data storage. All components were deployed directly from GitHub, enabling continuous integration and easy updates. This approach ensures high availability, performance, and simplified maintenance.



## 5.CONCLUSION

The **Periyar University NSS Portal** is a comprehensive web-based solution designed to modernize and streamline the management of National Service Scheme activities. It replaces traditional manual and paper-based processes with a centralized digital platform, enabling efficient handling of volunteer enrollment, event management, attendance tracking, and report generation. This significantly reduces manual effort, minimizes errors, and improves overall data accuracy and management.

The system follows a role-based access structure, allowing Admin, College, and Unit users to perform their responsibilities in a secure and organized manner. It is structured into key modules such as Authentication & Secure Access, Administrator Command, College Administration, Unit Operations & Volunteer Management, Student Enrolment & Demographic, Automated Document & Reporting, and Event Management & Field Activity, ensuring efficient system functionality.

The application is developed using modern technologies such as React for the frontend, Node.js and Express for the backend, and MongoDB Atlas for database management, providing a scalable and high-performance environment. The platform also enhances communication through automated notifications and offers a user-friendly interface with structured dashboards.

Overall, the system improves transparency, coordination, and efficiency, making it a reliable and future-ready solution for managing NSS activities.

## 6.BIBLIOGRAPHY

### Web References

- Official MongoDB Documentation – <https://www.mongodb.com/docs/>
- React Documentation – <https://react.dev/>
- Node.js Documentation – <https://nodejs.org/en/docs/>
- Express.js Documentation – <https://expressjs.com/>
- Mongoose Documentation – <https://mongoosejs.com/docs/>
- Chart.js Documentation – <https://www.chartjs.org/docs/>
- Netlify Documentation – <https://docs.netlify.com/>
- Render Documentation – <https://render.com/docs/>
- MongoDB Atlas Documentation – <https://www.mongodb.com/atlas/docs/>
- GitHub Documentation – <https://docs.github.com/>

### Book References

- Web Development with Node and Express – Ethan Brown
- Learning React – Alex Banks & Eve Porcello
- MongoDB: The Definitive Guide – Kristina Chodorow
- JavaScript: The Good Parts – Douglas Crockford
- Eloquent JavaScript – Marijn Haverbeke



## **A.DATA FLOW DIAGRAM**

## B.TABLE STRUCTURE

### MEMBER TABLE

| Field Name      | Type            | Description          |
|-----------------|-----------------|----------------------|
| name            | String          | Member name          |
| regNo           | String          | Register number      |
| dept            | String          | Department           |
| course          | String          | Course name          |
| community       | String          | Community category   |
| contact         | String          | Phone number         |
| bloodGroup      | String          | Blood group          |
| dob             | String          | Date of birth        |
| batchFrom       | String          | Batch start year     |
| batchTo         | String          | Batch end year       |
| unitId          | ObjectId (Unit) | Reference to Unit    |
| collegeld       | ObjectId (User) | Reference to College |
| fatherName      | String          | Father name          |
| fatherPhone     | String          | Father contact       |
| sex             | String          | Gender               |
| height          | String          | Height               |
| weight          | String          | Weight               |
| aadhaar         | String          | Aadhaar number       |
| enrolmentDate   | String          | NSS enrolment date   |
| culturalTalents | String          | Talents              |
| hobbies         | String          | Hobbies              |
| address         | String          | Address              |
| enrolmentNo     | String          | NSS enrolment number |
| remarks         | String          | Remarks              |
| universityName  | String          | University name      |
| email           | String          | Email                |
| isEnrolled      | Boolean         | Enrolment status     |

## UNIT TABLE

| Field Name  | Type                   | Description       |
|-------------|------------------------|-------------------|
| name        | String                 | Unit name         |
| head        | Mixed (ProgramOfficer) | Unit head         |
| contact     | String                 | Contact number    |
| mail        | String                 | Email             |
| password    | String                 | Hashed password   |
| members     | ObjectId[] (Member)    | List of members   |
| unitNumber  | String                 | NSS unit number   |
| createdDate | String                 | Created date      |
| events      | Array                  | Events list       |
| collegeId   | ObjectId (User)        | College reference |

## USER (COLLEGE) TABLE

| Field Name       | Type              | Description         |
|------------------|-------------------|---------------------|
| userName         | String            | Login username      |
| password         | String            | Hashed password     |
| insName          | String            | Institution name    |
| location         | String            | Location            |
| block            | String            | Block               |
| taluk            | String            | Taluk               |
| district         | String            | District            |
| pincode          | String            | Pincode             |
| universityName   | String            | University          |
| events           | Array             | Events              |
| code             | String            | Unique code         |
| units            | ObjectId[] (Unit) | Units under college |
| adoptingVillages | Array             | Village details     |

## PROGRAM OFFICER TABLE

| Field Name         | Type            | Description       |
|--------------------|-----------------|-------------------|
| name               | String          | Officer name      |
| designation        | String          | Designation       |
| department         | String          | Department        |
| unit               | String          | Unit              |
| college            | String          | College           |
| image              | String          | Photo             |
| dob                | String          | DOB               |
| community          | String          | Category          |
| email              | String          | Email             |
| mobile             | String          | Phone             |
| address            | String          | Address           |
| block              | String          | Block             |
| taluk              | String          | Taluk             |
| district           | String          | District          |
| pincode            | String          | Pincode           |
| dateOfAppointment  | String          | Appointment date  |
| teachingExperience | String          | Experience        |
| qualification      | String          | Qualification     |
| seminars           | String[]        | Seminars          |
| etiCompleted       | String          | ETI completed     |
| nssExperience      | String[]        | NSS experience    |
| specialTalent      | String[]        | Talents           |
| collegeId          | ObjectId (User) | College reference |
| officerID          | String          | Unique ID         |
| createdAt          | Date            | Created date      |

## EVENT TABLE

| Field Name  | Type                | Description       |
|-------------|---------------------|-------------------|
| name        | String              | Event name        |
| description | String              | Description       |
| category    | String              | Event type        |
| singleDay   | Boolean             | Single/multi-day  |
| date        | String              | Single day date   |
| dateFrom    | String              | Start date        |
| dateTo      | String              | End date          |
| timeFrom    | String              | Start time        |
| timeTo      | String              | End time          |
| venue       | String              | Venue             |
| images      | Array               | Image URLs        |
| brochure    | String              | Brochure file     |
| eventCode   | String              | Unique code       |
| unitId      | ObjectId (Unit)     | Unit reference    |
| attendees   | ObjectId[] (Member) | Participants      |
| collegeId   | ObjectId (User)     | College reference |

## Report Sub-Table

| Field Name        | Type    |
|-------------------|---------|
| conductedOnDate   | Boolean |
| participantsCount | Number  |
| collegesCount     | Number  |
| outcome           | String  |
| reportFile        | String  |
| reportPhotos      | Array   |
| submittedAt       | Date    |

## Adopting Villages Sub-Table

| Field Name | Type   |
|------------|--------|
| name       | String |
| address    | String |
| block      | String |
| taluk      | String |
| district   | String |
| pincode    | String |

## C.SAMPLE CODING

```
const express = require('express');

const mongoose = require('mongoose');

require('dotenv').config();

const jwt = require('jsonwebtoken');

const cors = require("cors");

const bcrypt = require('bcryptjs');

const verifyToken = require('./middleware/auth');

if (process.env.BREVO_API_KEY) {

  console.log(' Brevo Email service is configured');

} else {

  console.warn(' BREVO_API_KEY is not set in environment variables');

}


const sendEmailViaBrevo = async ({ from, to, subject, html, bcc }) => {

  const formatEmail = (email) => {

    if (typeof email === 'object') return email;

    return { email };

  };

  const toArr = Array.isArray(to) ? to.map(formatEmail) : (to ? [{ email: to }] : []);
```

```

const bccArr = Array.isArray(bcc) ? bcc.map(formatEmail) : (bcc ? [{ email: bcc }] : []);

const payload = {
  sender: formatEmail(from || process.env.EMAIL_USER || 'noreply@example.com'),
  subject: subject,
  htmlContent: html,
};

if (toArr.length > 0) payload.to = toArr;

if (toArr.length === 0 && bccArr.length > 0) {
  payload.to = [formatEmail(from || process.env.EMAIL_USER ||
'noreply@example.com')];
}

if (bccArr.length > 0) payload.bcc = bccArr;

const response = await fetch('https://api.brevo.com/v3/smtp/email', {
  method: 'POST',
  headers: {
    'Accept': 'application/json',
    'api-key': process.env.BREVO_API_KEY,

```

```

        'Content-Type': 'application/json'

    },

    body: JSON.stringify(payload)

});

if (!response.ok) {

    const contentType = response.headers.get('content-type');

    let errData = await response.text();

    if (contentType && contentType.includes('application/json')) {

        try { errData = JSON.parse(errData); } catch (e) { }

    }

    const error = new Error(`Brevo API Error: ${typeof errData === 'object' ?
JSON.stringify(errData) : errData}`);

    error.code = response.status;

    throw error;

}

const data = await response.json();

return { response: '250 Message queued', messageId: data.messageId || 'unknown' };

app.listen(PORT, () => {

    console.log(`Server is running on http://localhost:${PORT}`);

});

```



## D.SAMPLE INPUT

### Add New NSS Volunteer

✕

📄 Basic Details

NAME \*

Sam

REGISTER NUMBER \*

U25PG507CAP021

DEPARTMENT \*

MCA

COURSE \*

CS

BATCH FROM (YEAR) \*

2025

BATCH TO (YEAR) \*

2027

COMMUNITY \*

BC

BLOOD GROUP \*

B+

DATE OF BIRTH \*

10/19/2004

📅

MOBILE PHONE NO. \*

987654321

📄 Enrolment Details

📄 Enrolment Details

SEX \*

Male

FATHER/G UARDIAN 'S NAME \*

J Benjamin Sagayaraj

FATHER/G UARDIAN 'S CONTACT \*

9952759536

HEIGHT(CM) \*

177

WEIGHT(KG) \*

84

AADHAAR NUMBER \*

54897764214

EMAIL ID \*

bsam53888@gmail.com

DATE OF ENROLMENT

04/10/2026

📅

CULTURAL TALENTS

Musician

HOBBIES

Playing Keyboard

UNIVERSITY NAME

Periyar University

RESIDENTIAL ADDRESS \*

9/197 Udhayam Nagar, Kamalapuram, Omalur, Salem.

# E.SAMPLE OUTPUT



Sample Institute

888 NSS Organization

Explore Events

Admin

Adopting Villages

1 Village(s) + Add Village

R.C.Chettipatty

In between omalur and kamalapuram

Block: OmalurTaluk: OmalurDist: Salem

636309

NSS Units Management

Manage your college NSS units and members

View Program Officers

View All Students

+ Create New Unit

Register Program Officer

unit1

PUNSSSAM88801

Head: officer3

Created: 2026-03-27

Members: 20013

Delete Unit

Unit 2

PUNSSSAM88802

Head: Officer3

Created: 2026-03-29

Members: 6

Delete Unit

unit1

PUNSSSAM88801

Sample Institute

College Dashboard

Unit Details

UNIT CODE

PUNSSSAM88801

UNIT HEAD

officer3

EMAIL

catchyourcourse@gmail.com

UNIT NAME

unit1

CONTACT

9876543210

Quick Actions

Create Events

Manage events, volunteers and more from here.

Explore Events

Explore events created by other units.

Unit Members

Manage student volunteers

Search members...

Excel Export

UPLOAD VIA EXCEL

+ Add Member

| <input type="checkbox"/> | S.NO | NAME         | REG NO  | DEPARTMENT | BATCH     | CONTACT    | ACTION                                                        |
|--------------------------|------|--------------|---------|------------|-----------|------------|---------------------------------------------------------------|
| <input type="checkbox"/> | 1    | KarthikeyanR | 22CS001 | CS         | 2022-2026 | 9876543210 | <div><div>✓ Edit</div><div>Delete</div><div>+ PDF</div></div> |
| <input type="checkbox"/> | 2    | PriyaS       | 22EC045 | MCA        | 2022-2026 | 9123456789 | <div><div>✓ Edit</div><div>Delete</div><div>+ PDF</div></div> |
| <input type="checkbox"/> | 3    | DineshKumarM | 22ME012 | MBA        | 2022-2026 | 9988776655 | <div><div>✓ Edit</div><div>Delete</div><div>+ PDF</div></div> |

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